



LAUNDRYLAB

Multi-Branch Digital Management Platform

TECHNICAL INFRASTRUCTURE & OPERATIONAL FINANCIAL ANALYSIS

Application Running Costs Proposal

A comprehensive operational cost breakdown for hosting and managing the LaundryLab platform across 6 laundromat branches (24 machines) and 500+ active customers.

TARGET SCOPE

6 Laundromat Branches

PREPARED FOR

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DATE

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1 Executive Summary & Operational Scale

The **LaundryLab** application is a custom multi-branch digital management system designed to streamline daily laundry operations, track washing machine jobs (4 machines per location across 6 branches), manage customer orders, and send automated order status updates.

Operating Scale 6 Locations: Absa Towers, Zuri, Nala, The Encore, Georgia, Centurion 24 Machines Total	Customer Base 50+ Active Customers per branch (~300 - 500 total active clients) ~30–100 daily orders combined	Resource Footprint Light Compute: <100k API requests/month Light Data: <500 MB DB size over 1–2 years
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2 Detailed Cost Options Breakdown (ZAR)

Below are the two primary deployment models for the LaundryLab platform. All figures are given in **South African Rand (ZAR)**.

OPTION 1: Budget-Optimized Free Cloud Stack (RECOMMENDED)

Utilizes cloud provider free tiers combined with a business-hours keep-alive ping to eliminate cold starts.

R89.00 / Year

Component	Service Provider	Plan Tier	Monthly (ZAR)	Annual (ZAR)
Frontend Hosting	Vercel / Cloudflare Pages	Hobby / Free Tier	R0.00	R0.00
Backend API	Render.com / Koyeb (Python)	Free Web Service + Keep-Alive Ping	R0.00	R0.00
Database Engine	Supabase / Neon PostgreSQL	Free Tier (500 MB DB space)	R0.00	R0.00
Customer WhatsApp Alerts	WhatsApp Integration	Direct `wa.me` links / Free Meta Tier	R0.00	R0.00
Custom Domain Name	xneelo / Domains.co.za	`.co.za` Registration & Renewal	~R7.42 / mo	R89.00 / yr
TOTAL RUNNING COST (OPTION 1)			~R7.42 / mo	R89.00 / year

OPTION 2: Dedicated Managed 24/7 Cloud Stack

Always-on dedicated cloud server with guaranteed 24/7 uptime without keep-alive pings.

**~R135.00 /
Month**

Component	Service Provider	Plan Tier	Monthly (ZAR)	Annual (ZAR)
Frontend Hosting	Vercel / Cloudflare Pages	Hobby / Free Tier	R0.00	R0.00
Backend API	Render / Railway (Python)	Starter Dedicated Compute (\$7/mo)	~R125.00 / mo	R1,500.00 / yr
Database Engine	Supabase PostgreSQL	Free / Managed Tier	R0.00	R0.00
Customer WhatsApp Alerts	WhatsApp Integration	Direct `wa.me` links / Free Meta Tier	R0.00	R0.00
Custom Domain Name	xneelo / Domains.co.za	`.co.za` Registration & Renewal	~R7.42 / mo	R89.00 / yr
TOTAL RUNNING COST (OPTION 2)			~R132.42 / mo	R1,589.00 / year

3 Cold Start Management & Mitigation Strategy

Understanding Cloud "Cold Starts"

On free cloud hosting tiers (such as Render Free Tier), if no staff member touches the application for 15 minutes, the cloud provider puts the backend process to sleep to save energy. The next request takes **30 to 45 seconds** to wake up the server.

How Option 1 Completely Eliminates Cold Starts (R0 Cost Solution):

Automated Business-Hours Keep-Alive Ping

We integrate a free automated ping service (e.g., *cron-job.org* or *UptimeRobot*) configured to send a lightweight background health check ping (GET /api/health) to the backend every **10 minutes** between **06:00 AM and 08:00 PM** daily.

- **Result:** Render sees activity every 10 minutes, keeping the server **100% awake** throughout operational hours.
- **User Experience:** Instant load times for staff and customers at all 6 laundromat branches.
- **Cost:** R0 / month.

4 WhatsApp Messaging Strategy & Zero Cost Breakdown

To notify customers when their laundry is ready for pickup without incurring SMS costs (which average ~35c per SMS), we implement WhatsApp messaging:

Staff One-Click WhatsApp Link (R0)

When an order status is marked as *"Completed / Ready for Pickup"*, the staff member clicks a **"Notify via WhatsApp"** button on the POS screen.

This instantly opens WhatsApp Web or the WhatsApp app on the staff phone/tablet with a pre-populated message:

"Hi [Customer Name], your laundry order #[ID] is ready for pickup at [Branch Name]!"

Meta WhatsApp API Free Tier (R0)

Meta (WhatsApp's parent company) provides **1,000 FREE Service/Utility Conversations per month** for automated messaging.

Across 6 laundromat branches handling ~500 customer notifications per month, all automated WhatsApp alerts fit completely inside Meta's free monthly allocation.

Final Recommendation

By utilizing Option 1 with our automated Keep-Alive ping and WhatsApp customer notification system, your total recurring operational cost to run the LaundryLab platform across all 6 laundromats is **just R89.00 ZAR per year** for the .co.za domain name.